Rafi Mahmud Tumon

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Education

Technical University Berlin Berlin, Germany
Master of Science in Geodesy and Geoinformation Science *October 2020 – Present*

- Specialization: Geoinformation Science; Key Coursework: Photogrammetric Computer Vision, Geoinformatics, Geodatabases and Infrastructures, Space Geodesy, Adjustment Theory, Deep Learning for Geographic Data

Rajshahi University of Engineering and Technology Rajshahi, Bangladesh
Bachelor of Science in Urban and Regional Planning *March 2014 – November 2018*

Work Experience

OXG Glasfaser GmbH Berlin, Germany
Working Student, Strategic Area Development *June 2024 – Present*

- Improved operational efficiency by 20% through automated geospatial analysis using QGIS and Python for fiber network deployment.
- Conduct cost-benefit analyses and site assessments to identify optimal deployment areas while navigating geographical constraints and market competition.
- Developed Python-based geocoding tool for German addresses, reducing processing time by 30%.

Department of Civil Engineering, RUET Rajshahi, Bangladesh
Project Assistant (GIS and Geospatial Planner) *February 2020 – April 2020*

- Led an airport master planning project using GIS analysis to improve regional connectivity for underserved communities in northwestern Bangladesh.
- Collaborated with engineering teams and stakeholders to integrate spatial analysis into transportation planning decisions.

Urban Development Directorate Rajshahi, Bangladesh
GIS Analyst *December 2018 – January 2020*

- Conducted socio-economic surveys and spatial analysis for an urban planning project in Bagha Upazilla, collecting field data from local communities.
- Maintained geodatabases tracking urbanization indicators to support resource allocation decisions for deprived areas.

Research Experience

Flood Vulnerability Assessment Using Geospatial Analysis *November 2018*
Bachelor's Thesis: "Application of Remote Sensing and GIS for Flood Inundation Mapping and Vulnerability Assessment: A Case Study of Bogra, Sirajganj, and Tangail Districts"

- Conducted vulnerability assessment for three flood-prone districts in Bangladesh using satellite imagery to identify at-risk communities.
- Developed geospatial methodologies and produced flood risk maps to support disaster planning.

Personal Projects

Forest Loss Analysis in Colombia's Indigenous Lands
github.com/rafimt/Deforestation-Analysis

- Analyzed 10 years of satellite images using Google Earth Engine and Python to track deforestation and its effects on vulnerable communities, revealing a 15% loss in tree cover.

Wildfire Data Analysis
github.com/rafimt/Wildfire_Analysis_2022

- Developed a Python tool to analyze wildfire patterns in 8 countries (analyzing 1,234 fires total) to help with disaster planning.

Berlin's Environmental Dynamics Analysis
github.com/rafimt/Project-Geoinformatics-2024

- Assessed Urban Heat Island effects using satellite data, finding that built-up areas experienced the highest temperatures compared to other land types.

Skills and Tools

- **Core Technical:** GIS, Remote Sensing, Spatial Analysis, Geodatabase Maintenance, Spatial Data Management, Field Research & Data Collection
- **QGIS:** Data Management, Coordinate Reference Systems, Map Creation and Cartography, Spatial Joins, Plugin Usage, Database Connectivity, Processing Modeler, Python Scripting and Workflow Automation
- **Programming:** Python (Pandas, Geopandas, Rasterio, OSMnx, Folium, Scikit-learn, Seaborn, Pytorch, Matplotlib), SQL, JavaScript
- **Tools & Software:** Google Earth Engine, Git, Jupyter Notebook, Excel, PostgreSQL/PostGIS, MATLAB
- **Web GIS:** Django, HTML, CSS, Leaflet, Bootstrap
- **Machine Learning:** Supervised/Unsupervised Learning, Neural Networks, Model Evaluation, Classification, Object Detection, CNN, Vision Transformer, Graph Neural Network
- **Geospatial Expertise:** Engineering Survey, Least Squares Adjustment, Cartographic Principles, GNSS Processing, Environmental Analysis, Vulnerability Assessment

Awards and Recognition

- **Planning and Design of Compact Township, 2nd Runner Up**
  Inter-University Undergraduate Planning Students Competition, Bangladesh University of Engineering and Technology (BUET)
- **Publication**
  "Identification of Determinant Factors Influencing Modal Choice Behavior and Satisfaction Level of Commuters: A Case of Rajshahi City"
  Trends in Civil Engineering and its Architecture, Lupine Publishers
  DOI: 10.32474/TCEIA.2019.03.000171

Languages

English (Professional working proficiency), **German** (Elementary proficiency - A2)